Dynamic Macroeconomics II: Midterm Exam

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ITAM

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Duration of the exam: 1:30 hours

Name:

Student ID number:

Read carefully:

1. The exam has a total of 100 points.

2. This is a multiple choice exam. Please choose your answers directly on the exam. Use pen, not pencil.

3. You may make calculations on the designated *color* sheets of paper that I will provide you. But no answer in those color sheets counts for your grade.

4. It is forbidden to use any kind of electronic device: calculators, cellphones, tablets, laptops or any other.

5. It is forbidden to use class material or your own notes in any format: paper or digital.

6. It is forbidden to copy the answers of a classmate.

1. Fiscal Policies, based on Ljungqvist and Sargent (2004), Exercise 11.2 (total points of this question: 30)

A. Utility function

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

$0 < \beta < 1, c_t$ is consumption. u is strictly concave, strictly increasing and twice continuously differentiable. Labor supply n_t is inelastic and equal to 1.

B. Intertemporal budget constraint of the consumer

$q_t - q_{t+1}(1-\delta) = r_{t+1}(1-\tau_k)$

$$\sum_{t=0}^{\infty} \{q_t [(1+\tau_{ct})c_t + k_{t+1} - (1-\delta)k_t]\} \leq \sum_{t=0}^{\infty} \{r_t k_t (1-\tau_{kt}) + w_t n_t\}$$

τ_{ct} is a consumption tax, w_t is the present-value real wage, r_t is the present-value payment to capital, q_t is the intertemporal price, τ_{kt} is a tax on the return to capital ownership.

C. Intertemporal budget constraint of the government: g_t is spending.

$$\sum_{t=0}^{\infty} q_t(\tau_{ct}c_t + \tau_{kt}r_t k_t) = \sum_{t=0}^{\infty} q_t g_t$$

D. Feasibility

$$f(k_t, n_t) = c_t + k_{t+1} - (1-\delta)k_t + g_t$$

$0 < \delta < 1$

E. Technology

$$y_t = f(k_t, n_t)$$

f is homogeneous of degree 1, has decreasing marginal returns which are strictly positive.

F. $k_0 > 0$ given.

G. The firm maximizes

$$\sum_{t=0}^{\infty} [q_t f(k_t, n_t) - r_t k_t - w_t n_t]$$

max $\sum_{t=0}^{\infty}\beta^t u(c_t)$ s.a $\sum_{t=0}^{\infty} q_t[(1+\tau_{ct})c_t + k_{t+1} + (1-\delta)k_t] \le \sum_{t=0}^{\infty}[r_t k_t(1-\tau_{kt}) + w_t n_t$

$\mathcal{L}: \sum[\beta^t u(c_t) + \lambda[r_t k_t(1-\tau_{kt}) + w_t n_t - q_t[(1+\tau_{ct})c_t + k_{t+1} + (1-\delta)k_t$

$c_t: \beta^t u'(c_t) - \lambda q_t(1+\tau_{ct}) = 0$

$k_{t+1}: \lambda r_{t+1}(1-\tau_{kt+1}) - \lambda q_t + \lambda q_{t+1}(1-\delta) =$

$u'(c_t)/\beta u'(c_{t+1}) = q_t/q_{t+1}\,(1+\tau_{ct})/1+\tau_{ct+1}$

$u'(c_t)/\beta u'(c_{t+1}) = (Pmgk_t(1-\tau_{kt+1}) + 1-\delta)\left(\frac{1+\tau_{ct}}{1+\tau_{ct+1}}\right)$

max $\sum_{t=0}^{\infty} q_t f(k_t, n_t) - r_t k_t - w_t n_t$

$k_t: q_t f'(k_t) - r_t = 0$

$\Rightarrow r_t/q_t = Pmgk_t$

$\to r_{t+1}/q_{t+1}(1-\tau_{kt+1}) - q_t/q_{t+1} + (1-\delta) = 0$

$Pmgk_t(1-\tau_{kt+1}) + 1-\delta = q_t/q_{t+1}$

en ss $r/q = Pmgk$

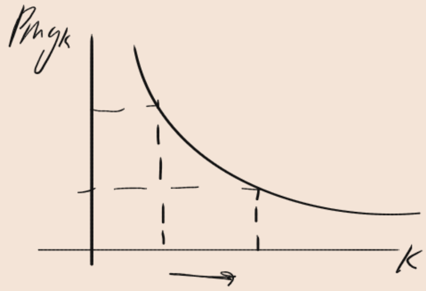
$f(k,1) = c - \delta k + g$

$r/q = Pmgk^*$

$1/\beta = Pmgk(1-\tau_k) + 1-\delta)$

$\Rightarrow Pmgk = 1/\beta + 1-\delta \big/ 1-\tau_k$

si $\tau_k = 0 \Rightarrow Pmgk^* = 1/\beta + 1-\delta$



1.1 Suppose that the economy has constant government spending $g_t = g > 0$, constant $1 > \tau_{kt} = \tau_k > 0$, and constant $\tau_c = 0$ for all t.

Calculate the steady state levels of capital, consumption and r_t/q_t. Go as far as getting one equation in one unknown for each variable. Notice I did not give you a specific $f(k_t, n_t)$. (Points: 15)

$k_{t+1} = I + (1-\delta)k_t$

R1 $1 = \beta(1 - \delta + f_k(1 - \tau_k)), f(k, 1) = c + \delta k + g, r/q = f_k$

R2 $1 = \beta(1 - \delta + f_k - \tau_k), f(k, 1) = c + \delta k + g, r/q = f_k$

R3 $1 = \beta(1 - \delta + f_k(1 - \tau_k)), f(k, 1) = c + (1 - \delta)k + g, r/q = f_k$

R4 $1 = \beta(1 - \delta + f_k(1 - \tau_k)), f(k, 1) = c + \delta k + g, r/q = f_k(1 - \tau_k)$

1.2 Now, suppose that the economy's initial capital is equal to the one you computed in 1.1. Assume the economy is in a steady state. Assume that completely unexpectedly, the government changes its policy. Taxes on the return to capital are ruled out, set equal to zero, in all periods. The government has to tax consumption. The consumption tax is always constant. This tax is a fraction strictly between 0 and 1. The level of government spending remains the same.

Tell me what happens with the new steady state levels of capital, consumption and r_t/q_t. Do they increase, decrease, or stay the same? (Points: 15)

R1 Before the change, $f_k = \frac{\frac{1}{\beta} - 1 + \delta}{1 - \tau_k}$. After the change $f_k = \frac{1}{\beta} - 1 + \delta$. Therefore capital is bigger after the change. Therefore r/q is bigger after the change.

R2 Before the change, $f_k = \frac{\frac{1}{\beta} - 1 + \delta}{1 - \tau_k}$. After the change $f_k = \frac{1}{\beta} - 1 + \delta$. Therefore capital is bigger after the change. Therefore r/q is smaller after the change.

R3 Before the change, $f_k = \frac{\frac{1}{\beta} - 1 + \delta}{1 - \tau_k}$. After the change $f_k = \frac{1}{\beta} - 1 + \delta$. Therefore capital is smaller after the change. Therefore r/q is larger after the change.

R4 Before the change, $f_k = \frac{\frac{1}{\beta} - 1 + \delta}{1 - \tau_k}$. After the change $f_k = \frac{1}{\beta} - 1 + \delta$. Therefore capital is smaller after the change. Therefore r/q is smaller after the change.

2. Model with borrowing constraints and an MIT shock (points: 30)

A. Intertemporal utility

$$\sum_{t=0}^{\infty} \beta^t u(c_t)$$

$0 < \beta < 1$. u is strictly increasing, strictly concave and twice differentiable. Besides, $\lim_{c \to 0} u_{ct} = +\infty$.

B. Financial market

There is a risk-free asset b_{t+1} that has a constant gross return $R > 1$. Assume $R\beta = 1$.

C. Sequential constraint of the consumer

The household receives an exogenous endowment y_t every period.

$$\begin{aligned} c_t + R^{-1} b_{t+1} &\leq y_t + b_t \\ y_t &\geq 0 \forall t \\ \sum_{t=0}^{\infty} \beta^t y_t &< +\infty \end{aligned}$$

$\beta^t u'(c_t) = \lambda_t$

$b_{t+1}: \lambda_{t+1} = \lambda_t R^{-1}$

$u'(c_t) = R\beta u'(c_{t+1})$

$\Rightarrow c_t = c_{t+1}$

$c_t + R^{-1} b_{t+1} \leq y_t + b_t$

$R^{-1} c_{t+1} + R^{-2} b_{t+2} \leq R^{-1} y_{t+1} + R^{-1} b_{t+1}$

D. Initial condition: $b_0 = 0$.

E. Transversality condition: $\lim_{T \to \infty} R^{-T} b_{t+T} = 0$.

$\sum R^{-t} c_t \leq \sum R^{-t} y_t$

$\Rightarrow c \sum R^{-t} \leq y \sum R^{-t}$

$\epsilon = R^{-1}$

$\Rightarrow c$.

$$\max \sum_{t=0}^{\infty} \beta^t u(c_t) \quad s.a \quad c_t + b_{t+1}/R \leq y_t + b_t \quad \forall t$$

$$\mathcal{L}: \sum \left[\beta^t u(c_t) + \lambda_t \left(y_t + b_t - c_t - b_{t+1}/R \right)\right]$$

$c_t: \beta^t u'(c_t) = \lambda_t \rightarrow \dfrac{u'(c_t)}{\beta u'(c_{t+1})} = R$ si $\beta R = 1$ ent $c_t = c \ \forall t$

$b_{t+1}: -\lambda_t/R + \lambda_{t+1} = 0 \Rightarrow \lambda_t/\lambda_{t+1} = R$

R.P.I

$$\sum \beta^t c_t = \sum \beta^t y_t$$

si $y_t = y \ \forall t$

$= c/1-\beta = y/1-\beta \Rightarrow c = y$

si $y_1 = y_\ell \quad y_t = y \ \forall t > 1$

$c/1-\beta = y_\ell + \beta y + \beta^2 y + \dots$

$= y_\ell + \beta y \left[1 + \beta + \beta^2 + \dots\right]$

$= y_\ell + \beta/1-\beta \ y$

$c = (1-\beta) y_\ell + \beta y$

si $y_t = y_\ell \ \forall t$

$c = y_\ell$

Assume that the income sequence is

$y_t = y > 0, \forall t \geq 0.$

Assume that there is NO borrowing constraint.

2.1 Solve for the value of consumption as a function of parameters and exogenous variables. (Points: 6)

R1 $c = 1/y$

R2 $c = (1-\beta)y$

R3 $c = y$

R4 $c = \beta y$

NOW, READ CAREFULLY: TIME GOES ON, AND THE ECONOMY ARRIVES TO $t = 1$. Assume that at $t = 1$, completely unexpectedly, the income sequence changes. Somebody named this kind of shock, in an environment with certainty, an MIT shock.

Consider two cases:

Case 1, "V-shaped recovery": At $t = 1$ income takes the value y_l, $0 < y_l < y$. Then, for $t \geq 2, y_t = y$.

Case 2, "L-shaped recovery": For $t \geq 1, y_t = y_l$.

2.2 Compute the sequence of consumption c_t for $t \geq 1$, in Case 1. (Points: 8)

R1 $c = \beta(y_l + y)$

R2 $c = (1-\beta)y_l + y$

R3 $c = (1-\beta)y_l + \beta y$

R4 $c = (1-\beta)y_l$

$c \frac{1}{1-\beta} \leq y_l + y \sum_{t=1}^{\infty} R^{-t} + b_1$ ← esto importa

acá lo hubiese tenido mal si b_1 o $b_0 \neq 0$

$c = \left(\frac{R-1}{R}\right) y_l + y/R$

$c = (1-\beta) y_l + y\beta$

2.3 Compute the sequence of consumption c_t for $t \geq 1$, in Case 2. (Points: 8)

R1 $c = \beta y_l$

$c\left(\frac{R}{R-1}\right) \leq$

R2 $c = (1-\beta))y_l$

R3 $c = y_l$

R4 $c = \frac{\beta}{1-\beta} y_l$

Notamos que

$c_2 < c_1$

$\Rightarrow y_l < (1-\beta)y_l + \beta y$

$\Rightarrow \beta y_l < \beta y$

$\Rightarrow y_l < y$ ✓

2.4 There is an economist that lives in the model. She/he does not observe the income sequence. She only knows that the recovery can be as in Case 1, or as in Case 2. I.e. she does not know the **persistence** of the shock. Tell the economist how to distinguish between the two cases by looking at consumption data. (Points: 8)

R1 If consumption is high after the shock, income follows an L-shaped recovery.

R2 If consumption is low after the shock, income follows an V-shaped recovery.

R3 The economist does not have enough information to determine the income path.

R4 If consumption is lower after the shock, income follows an L-shaped recovery.

3. Optimal taxation, based on Ljungqvist and Sargent (2004), Exercise 15.7 (total points of this question: 40)

A. Utility function

$$\sum_{t=0}^{\infty} \beta^t u(c_t, l_t)$$

$0 < \beta < 1, c_t$ is consumption, l_t is leisure. u is strictly concave, strictly increasing in consumption and leisure and twice continuously differentiable.

B. Time constraint of the consumer

$$1 = l_t + n_t$$

where n_t is labor.

C. Sequential budget constraint of the consumer

$$c_t + k_{t+1} + \frac{b_{t+1}}{R_t} - b_t \leq (1 - \tau_{nt}) w_t n_t + (1 - \tau_{at})(r_t + 1 - \delta) k_t$$

τ_{nt} is a labor income tax, w_t is the real wage, r_t is the payment to capital, τ_{at} is a tax on capital earnings plus the asset value of capital net of depreciation. $b_0 = 0$.

D. Technology: depends on labor and capital.

$$y_t = f(k_t, n_t)$$

where f has constant returns to scale and decreasing and strictly positive marginal returns.

E. Feasibility

$$f(k_t, n_t) = c_t + k_{t+1} - (1 - \delta) k_t + g_t$$

F. The firm maximizes every period

$$f(k_t, n_t) - r_t k_t - w_t n_t$$

$r_t = Pmg_{k_t}\ q_t$

$W_t = Pmg_{n_t}\ q_t$

$$\max \sum \beta^t u(c_t, l_t) \quad s.a \quad c_t + k_{t+1} + b_{t+1}/R - b_t \le (1-\tau_{nt}) w_t n_t + (1-\tau_{at})(r_t + 1 - \delta) k_t$$

$$\mathcal{L} : \sum \left[\beta^t u(c_t, \overset{(1-n_t)}{l_t}) + \lambda_t [(1-\tau_{nt}) w_t n_t + (1-\tau_{at})(r_t+1-\delta) k_t - c_t - k_{t+1} - b_{t+1}/R + b_t] \right]$$

$$c_t : \beta^t u'_{ct} = \lambda_t \Rightarrow \frac{u_{ct}}{\beta u_{ct+1}} = R$$

$$k_{t+1} = (1-\tau_{at+1})(r_{t+1}+1-\delta)\lambda_{t+1} - \lambda_t = 0$$

$$n_t : -\beta^t u_{nt} + \lambda_t (1-\tau_{nt}) w_t = 0 \Rightarrow \frac{u_{nt}}{u_{ct}} = (1-\tau_{nt}) w_t$$

$$R = (1-\tau_{at+1})(r_{t+1}+1-\delta)$$

$$b_{t+1} : -\lambda_t/R + \lambda_{t+1} = 0$$

RPI $\sum q_t c_t = \sum (1-\tau_{nt}) w_t n_t + (1-\tau_{at})(r_t+1-\delta) k_t$

CPO ahora ver

$$c_t : \beta^t u_{ct} - \lambda q_t = 0$$

$$n_t : -\beta^t u_{nt} + \lambda (1-\tau_{nt}) w_t = 0$$

$$\sum \beta^t u_{ct}/\lambda \, c_t = \sum \beta^t u_{nt}/\lambda \, n_t + A$$

$$\sum \beta^t u_{ct} c_t = \sum \beta^t u_{nt} n_t + \lambda A \quad CI$$

$$V_t \equiv u(c_t, 1-n_t) + \phi [\beta^t u_{ct} c_t - \beta^t u_{nt} n_t]$$

$$\max \sum \beta^t V_t \quad s.a \quad f(k_t, n_t) = c_t + k_{t+1} - (1-\delta) k_t + g_t$$

$$J = \sum \beta^t \left[V_t + \theta_t [f(k_t, n_t) - c_t - k_{t+1} + (1-\delta) k_t - g_t] - \phi A \right]$$

$$k_{t+1} : \beta^{t+1} \theta_{t+1} f_{k_{t+1}} - \beta^t \theta_t + \beta^{t+1} \theta_{t+1} (1-\delta)$$

$$\beta^{t+1} \theta_{t+1} (f_{k_{t+1}} + 1 - \delta) - \beta^t \theta_t = 0 \quad \text{en SS} \quad f_k + 1 - \delta = 1/\beta$$

Euler $\dfrac{u_{ct}}{\beta u_{ct+1}} = (1-\tau_{at+1})(r_{t+1} + 1 - \delta)$

SS $1/\beta = (1-\tau_a)(r + 1 - \delta)$

$$f_k + 1 - \delta) = (1-\tau_a)(\overset{f_k}{r} + 1 - \delta)$$

In what follows, you can use the notation $q_t = \prod_{i=0}^{t-1} R_i^{-1}$ for the inverse of the product of interest rates. Assume $q_0 = 1$.

3.1 Find the No Arbitrage Condition. (Points: 10)

R1 $R_{t-1} = (1 - \tau_{at})(r_{t+1} + 1 - \delta)$

R2 $R_t = (1 - \tau_{at+1})r_{t+1} + 1 - \delta$

R3 $R_t = (1 - \tau_{at+1})(r_{t+1} - \delta)$

R4 $R_t = (1 - \tau_{at+1})(r_{t+1} + 1 - \delta)$

3.2 Find the Implementability condition. In this notation λ is the Lagrange multiplier on the intertemporal budget constraint. (Points: 10)

R1 $\sum_{t=0}^{\infty} \beta^t u_{ct} c_t = \sum_{t=0}^{\infty} \beta^t u_{nt} n_t + (1 - \tau_{a0})(r_0 + 1 - \delta)k_0$

R2 $\sum_{t=0}^{\infty} \beta^t u_{ct} c_t + \sum_{t=0}^{\infty} \beta^t u_{nt} n_t + (1 - \tau_{a0})(r_0 + 1 - \delta)k_0 \lambda = 0$

R3 $\sum_{t=0}^{\infty} \beta^t u_{ct} c_t = \sum_{t=0}^{\infty} \beta^t u_{nt} n_t + (1 - \tau_{a0})(r_0 + 1 - \delta)k_0 \lambda$

R4 $\sum_{t=0}^{\infty} \beta^t u_{ct} c_t = \sum_{t=0}^{\infty} \beta^t u_{ct} u_{nt} n_t + (1 - \tau_{a0})(r_0 + 1 - \delta)k_0 \lambda$

3.3 Define a function V_t as in the Primal Approach as seen in class. Take the derivative of the Ramsey Lagrangean with respect to k_{t+1}. Use θ_t as the multiplier for feasibility at time t. You get: (Points: 10)

R1 $\beta^t \theta_t + \beta^{t+1} \theta_{t+1}(f_{kt+1} + 1 - \delta) = 0$

R2 $\beta^t \theta_t(-) + \beta^{t+1} \theta_{t+1}(f_{kt+1} + 1 - \delta) = 0$

R3 $\beta^t \theta_t(-) + \beta^{t+1} \theta_{t+1}(f_{kt+1} - \delta) = 0$

R4 $\theta_t(-) + \beta^{-1} \theta_{t+1}(f_{kt+1} + 1 - \delta) = 0$

3.4 Assume there exists a unique steady state. Find an equation that gives the answer to what is the optimal value of the tax. (Points: 10)

R1 $\beta(f_k + 1 - \delta) = \beta \tau_a (f_k + 1 - \delta)$

R2 $\beta f_k = \beta(1 - \tau_a) f_k$

R3 $\beta(f_k + 1 - \delta) + \beta(1 - \tau_a)(f_k + 1 - \delta) = 0$

R4 $\beta(f_k + 1 - \delta) = \beta(1 - \tau_a)(f_k + 1 - \delta)$

1.1 Suppose that the economy has constant government spending $g_t = g > 0$, constant $1 > \tau_{kt} = \tau_k > 0$, and constant $\tau_c = 0$ for all t.

$$PV(C) \leq PV(wn) + \sum^{\infty} q_t((1-\delta)k_t - k_{t+1}) + \sum r_t k_t (1-\tau_k)$$

$$q_{t+1}(1-\delta)k_{t+1} - q_t k_{t+1} + r_{t+1}k_{t+1}(1-\tau_k)$$
$$k_{t+1}[q_{t+1}(1-\delta) - q_t + r_{t+1}(1-\tau_K)]$$

$\Rightarrow$ N.A.C.: $q_t = q_{t+1}(1-\delta) + r_{t+1}(1-\tau_K)$

$$\mathcal{L} = \sum^{\infty} \beta^t v(c_t) + \lambda(VP(\text{after tax income}) - VP(\text{gross I}) - \sum q_t c_t (1+\tau_c))$$

$k_t =$

c_t: $\beta^t v'(c_t) = \lambda q_t(1+\tau_c)$

$$\Rightarrow q_t = \frac{\beta^t v'(c_t)}{\lambda(1+\tau_c)}$$

$$\Rightarrow \text{Euler: } \frac{\beta^t v'(c_t)}{\lambda(1+\tau_{ct})} = \frac{\beta^{t+1}v'(c_{t+1})}{\lambda(1+\tau_{ct+1})}(1-\delta) + r_{t+1}(1-\tau_K)$$

$$\max \sum^{\infty} \pi_t$$

r_t: $q_t f_K(t) = r_t$

$$\Rightarrow SS: \frac{v'(t)}{\lambda(1+\tau_{ct})} = \frac{\beta v'(t+1)}{\lambda(1+\tau_{ct+1})}(1-\delta) + \frac{\beta v'(t+1)}{\lambda(1+\tau_{ct+1})} f_K(t)(1-\tau_K)$$

$$\frac{v'(t)}{(1+\tau_c)} = \frac{\beta v'(t+1)}{(1+\tau_c)}[(1-\delta) + f_K(t)(1-\tau_K)]$$

$$\Rightarrow 1 = \beta(1-\delta + \overbrace{f_K(1-\tau_K)}^{q})$$

$f(k,1) = c + \delta k + g$

$r/q = f_K$

3

NAC: $\frac{(1-\tau_{at+1})(r_{t+1}+1-\delta)}{R_t} - - = 0 \quad \forall t$

RPI: $\sum q_t(\cdot) = \sum q_t w_t n_t (1-\tau_{nt}) + k_0(1-\tau_{a0})(r_0+1-\delta)$

$$\mathcal{L} = \sum_{t=0}^{\infty} \beta^t u(t) + \lambda(VP(I) - VP(c))$$

c_t: $\beta^t u_c(t) = \lambda q_t$

l_t: $\beta^t u_n(t) = \lambda q_t w_t (1-\tau_{nt})$

Sustituyendo las dos C.P.O.'s en RPI

$$\Rightarrow \sum_{t=0}^{\infty} \frac{\beta^t u_c(t)c_t}{\lambda} = \sum_{t=0}^{\infty} \frac{\beta^t u_n(t)}{\lambda} n_t + k_0(1-\tau_{a0})(r_0+1-\delta)$$

$\hookrightarrow$ IC

Sea $A \equiv (1-\tau_{a0})(r_0+1-\delta)\lambda k_0$

$$\sum^{\infty} \beta^t u_c(t)c_t = \sum^{\infty} \beta^t u_n(t)n_t + A$$

Sea $V_t \equiv u(c_t, 1-n_t) + \Phi[u_c(t)c_t - u_n(t)n_t] - \Phi A$

$$J = \sum^{\infty} \beta^t \{V_t + \theta_t[f(k_t,n_t) - c_t - k_{t+1} + (1-\delta)k_t - g_t]\}$$

k_{t+1}: $-\beta^t\theta_t + \beta^{t+1}\theta_{t+1}(f_K(t+1)+1-\delta) = 0$

Notando que $q_{t+1}/q_t = \frac{1}{R_t}$

Euler: $\beta^t u_{ct} = \beta^{t+1}u_{ct+1}\, q_t/q_{t+1}$

$u_{ct} = \beta u_{ct+1} R_t$

$= \beta u_{ct+1}(1-\tau_{at+1})(r_{t+1}+1-\delta)$

En ss $\Rightarrow$ Eu: $1 = \beta(1-\tau_a)(r+1-\delta)$

$\Rightarrow$ Ram: $1 = \beta(f_K + (1-\delta)) \to f_K = r$

$\therefore$ La única manera en que son iguales es cuando $1-\tau_a = 0$